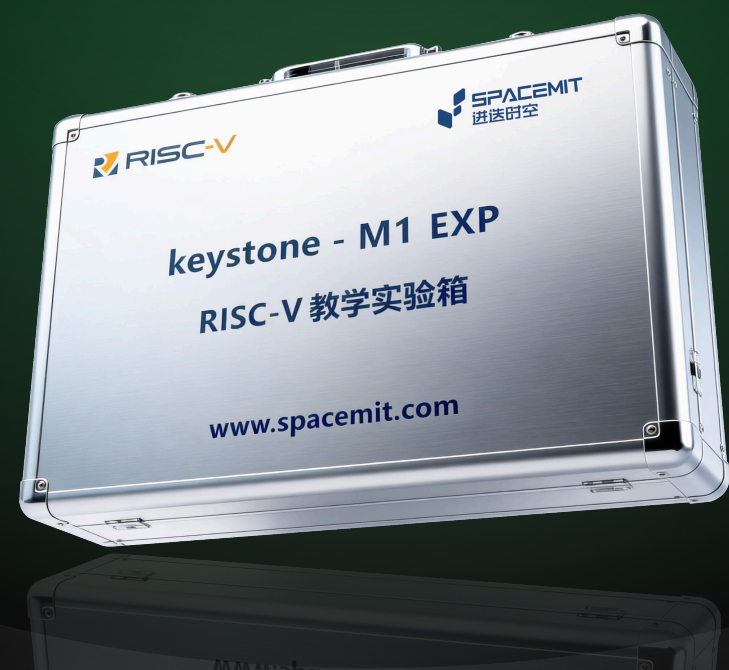


K1 Lab Kit Brief

RISC-V Architecture For Intelligent Future



A RISC-V Lab Kit for Embedded and AI Education

The RISC-V Lab Kit is designed for education in embedded systems, AI, and RISC-V computing. It combines curriculum-oriented hardware and software with RISC-teaching materials and hands-on labyguides, helping educators deliver practical courses and enabling students to build engineering skills.



CPU-Based AI Compute for Education

Provides AI inference capability in a CPU form factor, supporting model training and experimentation from 0.5B to 8B parameters



Modular, Detachable Design

Composed of a core board, baseboard, 10.1-inch touchscreen, and accessories for flexible configuration and cost efficiency



Plug-and-Play Peripheral Modules

Supports sensors, motors, fingerprint modules, robotic arms, and more for diverse lab scenarios



Comprehensive Teaching Resources

Includes 100+ lab examples and detailed manuals covering AI, signal processing, and measurement & control

Specifications

Processor	SpacemiT M1 with 2.0 TOPS AI compute
Display	HDMI Type-A, up to 1080p@60 Hz; integrated 10.1-inch IP Stouchscreen, 1920x1200
Memory	LPDDR4X with a 2400 MT/s speed, available in 16 GB
Storage	eMMC 5.1, 64 GB
Expansion	M.2 2280 M-Key for NVMe SSDs, up to 1 TB; TF card slot with UHS-II support
Wireless	Wi-Fi 6 & BT 5.2
Ethernet	Dual RJ45 Gigabit Ethernet ports, compatible with 1000M / 100M
Audio	3.5 mm headset jack
USB	4 × USB 3.0 Type-A Host ports
Module Slots	4 slots for GPIO, SPI, UART & I2C modules, supporting 30+ experiment modules
MIPI	2 × 4-lane MIPI CSI, supports "4 + 4" or "4 + 2 + 2" configurations
Appearance	Portable flip-top aluminum alloy case, 480 × 370 × 170 mm
OS	Bianbu, Ubuntu, OpenKylin, Deepin, Fedora
Power Input	12 V / 3 A adapter supply
Reliability	Acrylic isolation protection; ESD tolerance up to ±8 kV (contact) and ±15 kV (air) in module expansion areas

